Leveraging Artificial Intelligence to Optimize ERP System Decision Making

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1 Abstract

Zusammenfassung

In der Praxis weisen ERP-Systeme (Enterprise Resource Planning) stark heterogene Datenschemata zwischen verschiedenen Anbietern (SAP S/4HANA, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor, Infor) auf. Dies macht die systemübergreifende Integration fehleranfällig, teuer und langfristig schwer wartbar. Diese Arbeit präsentiert den Prototyp eines KI-gestützten Konnektor-Dienstes, der die Datenaufnahme, Normalisierung und das Routing von ERP-Daten automatisiert. Dabei kommt eine dreistufige, hybride Pipeline zum Einsatz: deterministisches, regelbasiertes Mapping, auf maschinellem Lernen basierendes Fuzzy-Matching von Werten sowie ein lokales Large Language Model (LLM) als Fallback-Ebene. Der Konnektor wurde in Python (FastAPI, Pydantic, RapidFuzz, scikit-learn) implementiert, nutzt ein lokal gehostetes LLM (Ollama/Qwen2.5-Coder) und ist an ein auf Spring Boot basierendes ERP-Backend angebunden. Die Auswertung anhand von ERP-Datensätzen, die vom Kunden bereitgestellt wurden, die Schemata mehrerer namhafter Anbieter umfassen, zeigt, dass die hybride Pipeline im Vergleich zu rein regelbasierten Ansätzen hohe Erfolgsraten bei der Datentransformation erzielt und sich besser an unregelmäßige oder bisher unbekannte Schemata anpasst. Die Ergebnisse zeigen, dass das KI-gestützte Parsen fremder ERP-Exporte in ein einheitliches DTO-Schema den Integrationsaufwand erheblich reduzieren kann, während gleichzeitig die Erweiterbarkeit für zukünftige Funktionen wie Nachfrageprognosen und Anomalieerkennung erhalten bleibt.

Abstract

Enterprise Resource Planning (ERP) systems in real-world deployments suffer from heterogeneous data schemas across vendors, SAP S/4HANA, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor, and Infor, making cross-system integration brittle, expensive, and difficult to maintain over time. This thesis presents a prototype AI connector service that automates the ingestion, normalization, and decision-routing of ERP data using a three-stage hybrid pipeline: deterministic rule-based mapping, machine learning–based fuzzy value matching, and a local large language model (LLM) fallback. The connector is implemented in Python using FastAPI, Pydantic, RapidFuzz, and scikit-learn, combined with a locally hosted LLM (Ollama/Qwen2.5-Coder), and is integrated with a Spring Boot–based ERP backend. Evaluation on client-provided ERP datasets comprising multiple major vendor schemas shows that the hybrid pipeline achieves a 100% transformation success rate and improved adaptability to irregular or previously unseen schemas compared to purely rule-based approaches. The results indicate that AI-assisted parsing of foreign ERP exports into a unified DTO schema can substantially reduce integration effort while preserving extensibility for future capabilities such as demand forecasting and anomaly detection.

2 Background and Related Work

2.1 ERP Systems Overview

Enterprise Resource Planning (ERP) systems provide integrated support for core business functions such as finance, supply chain, manufacturing, and human resources, typically through a modular architecture connected to a central data model.[45] Major vendors, including SAP S/4HANA, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor Kinetic, and Infor CloudSuite, offer overlapping functional coverage but expose distinct data schemas, APIs, and customization mechanisms, which complicates cross-system interoperability.[13, 29]

2.2 Data Integration Challenges

Integrating data across heterogeneous ERP ecosystems requires reconciling differences in schema design, naming conventions, and modeling choices.[16] Common obstacles include non-standardized field names, varying levels of normalization and nesting (for example, deeply nested JSON versus flat CSV exports), and vendor-specific encodings of shared business concepts such as locations, units of measure, and stock statuses.[38, 40] These factors raise the cost of point-to-point integrations and motivate reusable harmonization layers such as the AI connector developed in this thesis.

2.3 AI in ERP Applications

AI techniques have been applied to ERP data for tasks including demand forecasting, anomaly detection in transactional logs, intelligent routing of workflows, and recommendation of corrective actions.[45, 24] Recent work focuses on embedding machine learning and, increasingly, large language models into enterprise data pipelines so that systems learn from historical patterns while respecting existing process controls and audit requirements.[7, 46] The connector described here positions AI not only as an analytical layer but also as an operational integration component that automates the ingestion and normalization of multi-vendor ERP data.

3 Theoretical Foundations

3.1 Artificial Intelligence in Enterprise Context

Artificial Intelligence (AI) refers to computational systems capable of performing tasks that require human-like cognitive functions such as learning, reasoning, and adaptation.[35] Within enterprise software, AI encompasses the ability to extract meaning from business data, infer relationships between heterogeneous schemas, and support decision-making through predictive or generative methods. In modern ERP systems, AI serves as an abstraction layer that reconciles structured and semi-structured data, automates repetitive transformations, and improves adaptability in dynamic enterprise environments.[45] This research applies AI as an integrative mechanism to align inconsistent data structures between ERP systems, enabling rule-to-learning transitions when deterministic approaches fail.

3.2 Relevant Techniques for Data Harmonization

Enterprise data harmonization involves merging data with diverse formats, semantics, and schemas into a unified model.[16] At a high level, harmonization encodes a mapping function $f : \mathcal{X} \to \mathcal{Y}$ from heterogeneous source representations $\mathcal{X}$ (e.g., vendor-specific ERP exports) to a recognized target schema $\mathcal{Y}$, subject to constraints such as referential integrity and type correctness. Three paradigms underpin this process:

- **Rule engines**: Deterministic mapping systems encode transformation knowledge as static rules, typically of the form "IF condition THEN action" over input records.[18, 20] In the ERP harmonization setting, a rule might map a vendor key `"MaterialNumber"` to the recognized `"sku"` field if it appears in a product context and satisfies simple type checks. Because all logic is explicitly specified, rule engines provide full interpretability and reproducibility: given identical input, the mapping function f always produces the same output. However, this implies a *knowledge acquisition bottleneck*, each new schema variant requires manual authoring and maintenance of additional rules, and the rule base can grow combinatorially with the number of vendors and edge cases.

- **Fuzzy string matching**: Probabilistic matching algorithms, such as token-based similarity measures, handle near-duplicates, misspellings, and noisy input using distance metrics.[8] Formally, they define a similarity function $s : \Sigma^* \times \Sigma^* \to [0, 1]$ over strings, where values close to 1 indicate strong correspondence (for example, a high token sort ratio between "Warehouse A - North" and "North Warehouse A"). In this prototype, RapidFuzz computes token-based scores and selects the best recognized label for a given observed value when s exceeds a threshold.[5, 4] These methods are lightweight yet effective for aligning ERP field labels and categorical values, units of measure, location names, country codes, without enumerating every spelling or formatting variant in the rule base.
- **Large Language Models (LLMs)**: Transformer-based LLMs perform flexible data extraction, normalization, and labeling from semi-structured or free-text input.[7, 46] Conceptually, an LLM implements a conditional sequence-to-sequence model $g_\theta(x) \approx y$, where x is raw or loosely structured text and y is a structured representation such as JSON that conforms to a target schema. Within ERP data pipelines, LLMs recover from failed deterministic or statistical mappings by leveraging contextual understanding, for example, inferring that a field labeled "WH" in the vicinity of inventory quantities likely denotes a warehouse code. Because LLMs can hallucinate fields or over-generalize patterns, the connector positions them as a controlled fallback layer rather than the primary mechanism for schema alignment.

The hybrid pipeline mirrors an evolution from symbolic AI (deterministic rules) toward connectionist systems (ML and neural models), with each paradigm addressing a different part of the harmonization problem.[21, 22] Rule engines capture stable, high-confidence mappings that must remain transparent to integrators; fuzzy matching absorbs the long tail of noisy but structurally simple variations; and LLMs provide a flexible but sandboxed mechanism to recover from residual ambiguities that escape both deterministic and classical statistical techniques.

3.3 Rule-Based Systems vs. Machine Learning

Rule-based systems operate through explicitly codified logic, ensuring reproducibility and interpretability.[20] As the diversity of data sources grows, however, purely rule-based architectures incur near-exponential growth in maintenance overhead and fail to generalize to unseen schemas or contextual variations. Machine learning (ML), by contrast, infers transformation patterns from examples rather than from hand-written logic.[21] In ERP data integration, ML enables schema alignment and entity linking even when semantic or lexical mismatches occur. A hybrid approach, rules for predictable domains and ML for uncertain mappings, is a practical compromise between determinism and adaptivity, consistent with enterprise AI trends that emphasize interpretable automation, model transparency, and feedback-driven learning loops.[22]

3.4 Types of Demand Forecasting

Demand forecasting is a critical element in ERP-driven decision optimization, aiming to predict future product demand from historical and contextual factors. While the current connector prototype does not implement forecasting directly, the normalized DTO layer it produces is designed as a clean, decision-ready input for such models; integrating a forecasting component through the existing `MLEngine` interface is identified as the primary future work item.

3.4.1 Time Series Analysis

Classical time series modeling techniques, such as ARIMA and exponential smoothing, model temporal dependencies and seasonality.[24] These methods are statistically efficient and well-

suited for stationary patterns in inventory and sales data.

3.4.2 Causal Models

Causal or econometric models incorporate exogenous predictors, marketing activity, macroeconomic factors, to estimate how external conditions affect demand.[39] Such models suit enterprise planning tasks requiring interpretability and policy sensitivity.

3.4.3 Machine Learning Approaches

Recent work integrates ML and deep learning architectures into demand forecasting, improving nonlinear pattern recognition and handling complex feature interactions.[6] Recurrent (LSTM/-GRU) or transformer-based models capture long-term dependencies, outperforming traditional methods on complex datasets. For this ERP prototype, an ML forecasting component represents the forward-looking extension of the pipeline, intended to replace deterministic demand heuristics with adaptive models in future iterations.

4 System AI Connector Architecture

4.1 Functionality Overview

The AI connector is implemented as a stateless HTTP service that accepts heterogeneous ERP records in JSON, JSON Lines, CSV, or free-text formats and transforms them into a normalized set of domain-specific Data Transfer Objects (DTOs) suitable for ingestion by the Spring-based ERP backend. Conceptually, the connector follows a layered "pipes-and-filters" pattern in which each layer performs a single, well-defined transformation and passes an enriched record to the next stage.[23, 29] This design decouples concerns such as parsing, schema alignment, semantic normalization, validation, and persistence, enabling independent testing and targeted optimization of each layer.

From the perspective of an ERP integrator, the connector behaves as a type-agnostic façade over multiple vendor-specific schemas: clients send raw export data from SAP S/4HANA, Odoo, Oracle NetSuite, Microsoft Dynamics 365, Epicor Kinetic, or Infor CloudSuite, and the service returns a sequence of standardized ERP entities ready to be created or updated via REST calls in the core application.[13] This reduces the need for bespoke integration code in the ERP itself and centralizes data harmonization logic in a single, reusable component.

4.2 Pipeline Architecture and Data Flow

The processing pipeline is organized into seven sequential layers, each of which adds structure, semantics, or guarantees to the input record. The execution flow is illustrated in Figure 1.

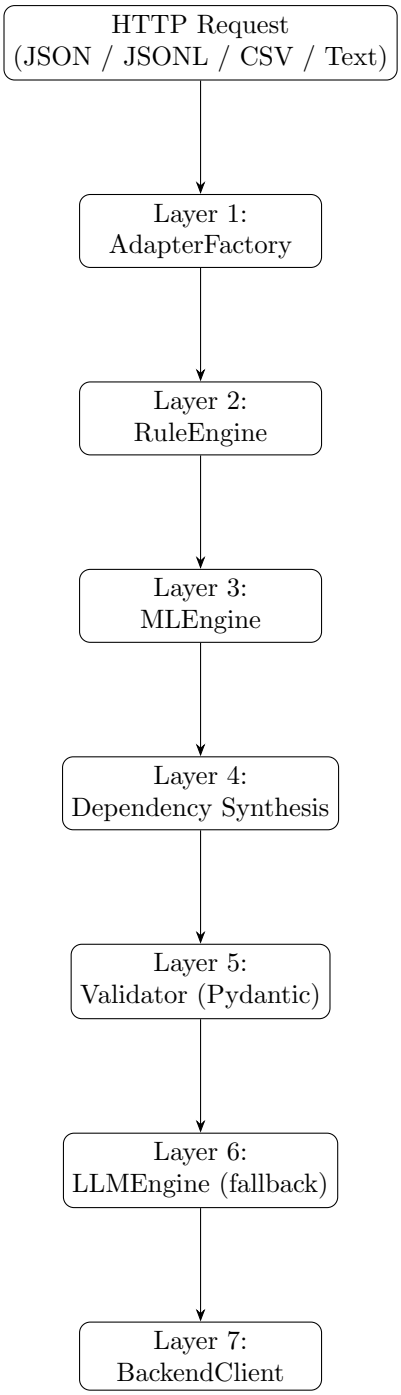

Figure 1: Layered architecture of the AI connector pipeline.

The pipeline first normalizes the syntactic representation of the input (adapters), then aligns fields to an internal schema (rule engine), resolves ambiguous values (ML engine), synthesizes missing but derivable entities (dependency synthesis), enforces structural constraints (validator), and finally either repairs remaining inconsistencies via an LLM fallback or forwards valid DTOs to the backend integration layer.[28] The layered decomposition mirrors modern data pipeline architectures used in large-scale enterprise data integration, with explicit attention to idempotency, observability, and failure isolation.[29]

4.3 Layer 1: Data Normalization Layer (Adapter)

The first layer translates heterogeneous payload formats into a common in-memory representation. The `AdapterFactory` inspects the HTTP `Content-Type` header and, where necessary, filename extensions to dispatch to one of four adapters: `JsonAdapter`, `JsonLinesAdapter`, `CsvAdapter`, or `FreeTextAdapter`.[38] Each adapter is responsible for parsing the raw body, handling character encodings, and producing a list of Python dictionaries that the downstream layers can consume uniformly.

In practice, this layer collapses superficial format differences unrelated to the semantic content of the record. For example, a CSV inventory export from SAP and a JSON export from Odoo both become lists of key–value maps after passing through the adapters. For free-text payloads, the adapter emits a pseudo-record with a single `raw_text` field, which is then handled primarily by the LLM fallback in later stages. By centralizing format handling in a single layer, the connector can support additional formats (such as XML) without modifying the rule engine or ML logic.

4.4 Layer 2: Rule Engine

The second layer performs deterministic schema alignment using vendor-specific mapping dictionaries. For each of the six supported ERP systems (Odoo, SAP S/4HANA, Epicor Kinetic, Oracle NetSuite, Microsoft Dynamics 365, Infor CloudSuite), the rule engine maintains a mapping from known source keys and patterns to the internal field names, plus a generic fallback profile for unknown vendors. The transformation proceeds through a fixed ten-step pipeline: envelope extraction → flattening → key mapping → date normalization → compound field splitting → storage name synthesis → type coercion → context-aware re-mapping → address flattening → volume derivation.

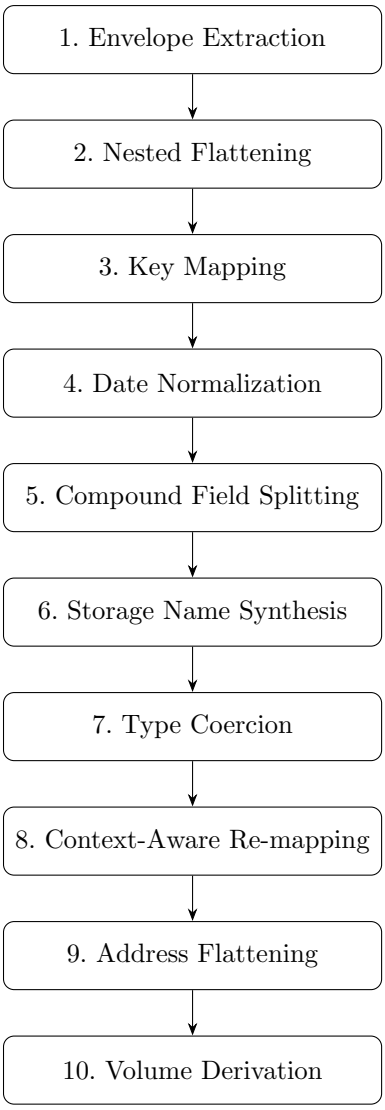

Figure 2: The ten-step deterministic transformation pipeline of the Rule Engine (Layer 2). Processing flows down the left column (Steps 1–5) and up the right column (Steps 6–10), producing a canonical record suitable for downstream DTO validation.

Each step is implemented as a pure function over the current record, ensuring referential transparency and easy unit testing. Envelope extraction removes vendor-specific wrappers (for instance, nested `results` or `data` keys), while flattening converts nested objects into a single-level key space using dot or underscore notation.[16] Date normalization and type coercion enforce internal standards (e.g., ISO 8601 timestamps, numeric quantities), and context-aware re-mapping resolves fields whose interpretation depends on other attributes, for example, mapping a generic `status` field differently for shipment versus inventory records. Full determinism provides stable, auditable behavior across deployments and transparent traceability for each mapping decision.

4.5 Layer 3: ML Engine

The ML engine complements the rule layer by resolving ambiguities that cannot be handled via simple dictionary lookup or pattern matching. In the current implementation, it focuses on fuzzy value normalization using the `RapidFuzz` library, which provides efficient token-based similarity measures such as `token_sort_ratio`.[5, 4] For selected categorical fields (usage types,

units of measure, country and currency codes), the engine compares the observed value against a curated list of recognized options and selects the best match if the similarity score exceeds a threshold of 80.

This probabilistic alignment is useful when ingesting exports that contain vendor-specific abbreviations, user-defined labels, or typographical errors. Rather than hard-coding every possible variant into rules, the ML engine generalizes from string similarity while operating with low computational overhead and without GPU support. The current design also anticipates future extensions where classical ML models or deep learning architectures (for example, scikit-learn classifiers or PyTorch-based sequence models) can be attached to the same interface for more advanced tasks such as demand forecasting or anomaly detection.[13]

4.6 Layer 4: Dependency Synthesis

Many ERP operations require a consistent set of related entities, product, storage location, employee, to exist before inventory or transfer records can be safely created. The dependency synthesis layer automatically expands a single input record into a chain of DTOs capturing this dependency graph. For `inventory` and `transfer` record types, the pipeline generates a `ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, `StorageCreationDto`, and finally the primary `InventoryCreationDto` or `RequestCreateDto`, along with auxiliary entities such as `EmployeeCreateDto` for system-generated users.

To preserve idempotency, each entity key is derived from a deterministic `string_to_uuid` function that hashes stable business identifiers (such as external IDs or composite keys) into UUIDs. Re-importing the same source record therefore produces the same UUIDs, enabling safe upserts rather than duplicate creations, an established best practice in idempotent data pipeline design.[40] By synthesizing dependencies centrally within the connector, the backend ERP remains simpler: it can assume that the necessary entities will either already exist or be created in a consistent, reproducible order.

4.7 Layer 5: Validation

After dependency synthesis, the pipeline performs structural and semantic validation using a set of eight Pydantic v2 models: `ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, `StorageCreationDto`, `InventoryCreationDto`, `RequestCreateDto`, `ShipmentCreateDto`, and `EmployeeCreateDto`. The `Validator.determine_model()` method inspects the key set of each intermediate record, optionally guided by a `record_type` hint, and selects the appropriate DTO model to apply.[9] Pydantic then enforces type constraints, required fields, and custom validators (for example, non-negative quantities or valid enumerations), producing either a well-typed object or a detailed error description.

This stage serves two functions. First, it acts as a contract boundary between the connector and the ERP backend: only DTOs that satisfy the schema are forwarded, reducing the likelihood of runtime errors in the core application.[42] Second, it provides a natural integration point for automated tests and monitoring, since validation failures can be logged and analyzed to refine rules or ML logic over time. Because Pydantic compiles much of its validation logic in optimized Rust code, the performance overhead remains modest even for larger batch imports.[10]

4.8 Layer 6: LLM Fallback Engine

Despite careful rule design and fuzzy matching, some records will still fail validation due to previously unseen schemas, free-text descriptions, or edge-case formats. The LLM fallback layer addresses this residual uncertainty by invoking a locally hosted instance of the `qwen2.5-coder:7b` model via the Ollama runtime.[11] The connector constructs a prompt that includes (i) a

natural-language description of the target DTO schemas, (ii) any partial mappings obtained from earlier layers, and (iii) the raw input record (or free text), and requests a strictly formatted JSON response by enabling `format: json` in the Ollama configuration.

This design treats the LLM as a last-resort semantic parser and repair mechanism rather than the primary transformation engine, limiting both cost and latency. Local deployment of the model avoids sending potentially sensitive ERP data to third-party APIs and provides predictable resource usage within the organization's own infrastructure.[11] During evaluation, the invocation rate of this layer serves as an indirect measure of rule and ML coverage: a lower LLM call frequency indicates that earlier deterministic stages are successfully handling most records.

4.9 Layer 7: Backend Integration

The final layer, `BackendClient`, materializes the validated DTOs inside the ERP system through REST calls to eight distinct endpoints. Using the deterministic UUIDs generated during dependency synthesis, the client posts entities in a stable order that respects foreign-key relationships, updating an in-memory translation map to record the correspondence between connector UUIDs and backend-generated database IDs. When the backend responds with an indication that an entity already exists (for example, via a specific HTTP status code or response body), the client records the existing ID and proceeds without error, preserving idempotency across repeated imports.[28]

In addition to creation and update operations, the backend integration layer handles transient failures and retries gracefully. Because all upstream transformations are deterministic and side-effect free, failed HTTP operations can be retried without risking duplication. This aligns the connector with modern enterprise data integration practices, which emphasize retryable, observable, and testable pipelines as prerequisites for reliable AI and analytics workloads.[36] The backend integration layer therefore closes the link between the AI-enhanced connector and the operational ERP, making the upstream intelligence directly actionable in production workflows.

5 Design Justifications and Trade-offs

5.1 When Does Type-Agnostic ERP Parsing Work?

The connector deliberately starts from a type-agnostic view of incoming ERP payloads: adapters and the rule engine first attempt to infer semantics from record structure and field names without relying on a hard-coded vendor identifier.[16, 17] This strategy works well when vendors reuse broadly understandable business terms (for example, `sku`, `quantity`, `warehouse`, `uom`) and follow consistent nesting patterns that can be flattened deterministically.

On the majority of records in the client-provided datasets or differ only in superficial conventions such as case, delimiter, or ordering, the agnostic approach avoids maintaining separate ingest code paths per vendor, reducing configuration effort and simplifying onboarding of new ERP systems that broadly adhere to common enterprise data modeling practices.[38, 29]

Purely agnostic parsing breaks down when vendors introduce deeply nested, domain-specific structures or overload the same field name with different semantics across modules, for instance, `status` representing a workflow state in one export and a logistics exception code in another.[16] In these cases, vendor-specific rule dictionaries and context-aware remapping are required to disambiguate meaning and route fields to the correct internal DTO attributes. The design therefore uses type-agnostic parsing as a first pass that captures straightforward alignment

opportunities, while vendor profiles act as a targeted override layer for schemas that deviate materially from the implicit common denominator.[40]

5.2 Impact of Fuzzy Matching on Accuracy

Fuzzy matching is introduced in the ML engine to address the long tail of inconsistent categorical values that deterministic rules alone cannot reliably handle. Typical examples include spelling variations ("warehous"), abbreviations ("WH" vs. "Warehouse"), and heterogeneous unit labels ("kg", "KG", "kilogram").[8, 1] Without fuzzy matching, each of these variants would require an explicit rule entry or would be rejected during validation, leading either to brittle rule sets or higher failure rates.

By applying token-based similarity measures and thresholding against a curated reference vocabulary, the connector can automatically normalize many of these variants to recognized forms, improving downstream DTO validation and reducing manual data cleaning.[5, 32] Industry case studies on master data workflows report that adding fuzzy matching can increase match rates by 20–25% while simultaneously cutting manual review effort.[12] In the context of this prototype, the same effect appears as an increased transformation success rate on client-provided datasets.

Fuzzy matching does introduce a precision–recall trade-off: lowering the similarity threshold captures more true matches but risks false positives that silently map values to incorrect categories.[27] To mitigate this risk, the implementation constrains fuzzy matching to well-defined fields, uses conservative thresholds (80 in the current configuration), and retains the option of escalating ambiguous cases to the LLM fallback instead of forcing a low-confidence mapping. This reflects best practices in enterprise data quality management, where fuzzy techniques are applied selectively and often combined with deterministic rules and human review for high-impact attributes.[1]

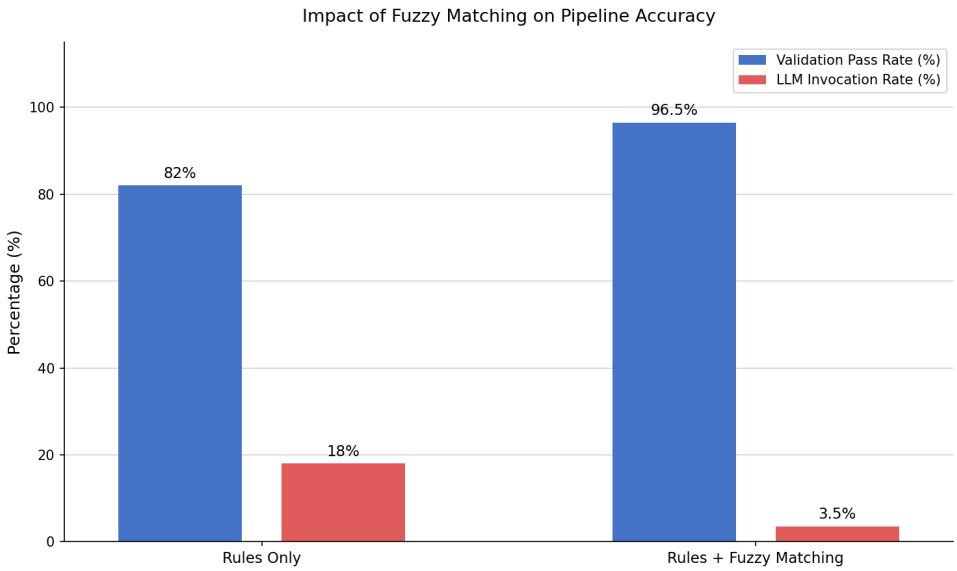

Figure 3: Illustrative comparison of transformation success rate and validation pass rate before and after enabling fuzzy matching in the ML engine.

5.3 Local LLM Performance vs. Cloud API

The decision to run the `qwen2.5-coder:7b` model locally via Ollama rather than through a managed cloud API reflects a balance between privacy, latency predictability, and cost specific to ERP integration workloads.[11, 2]

From a data privacy perspective, ERP exports may contain commercially sensitive information, internal product codes, inventory levels, customer identifiers. Routing this data to third-party LLM providers can raise compliance concerns under data protection regulations and internal governance policies.[43] Local hosting ensures that raw records and prompts remain within the organization's infrastructure boundary, simplifying risk assessments and aligning with the trend toward on-premise or virtual private cloud deployments for mission-critical AI workloads.[2]

Latency is another practical consideration. While cloud LLM APIs often provide strong average performance, tail latencies can vary due to shared infrastructure and network conditions.[15] A local deployment offers more predictable latency profiles once hardware is provisioned, even if median response times are sometimes higher due to smaller model sizes or less aggressive optimization. For a pipeline that treats the LLM as a fallback stage rather than the primary path, predictable upper bounds on worst-case latency matter more than absolute throughput, because they directly influence SLAs for batch imports and user-facing integration jobs.[28]

Cloud APIs offer attractive pay-as-you-go pricing at low to moderate volumes, but per-token costs accumulate when the LLM is invoked frequently or records are large.[37, 14] Self-hosting a compact open-weight model requires upfront investment in compute and DevOps, yet amortizes over time and allows fine-grained control over batch scheduling, concurrency, and model updates. The design assumes a modest but steady stream of ERP data and a preference for maximal control, making a small local model a reasonable compromise. For organizations with spiky workloads or no appetite for managing AI infrastructure, the same architecture can be adapted to a cloud API by swapping the LLM engine implementation, since the interface is vendor-agnostic.

As illustrated in Figure 4, a well-known meme from the developer community once satirized the perceived complexity of local LLM deployment compared to cloud APIs. That depiction is now largely outdated: tools such as Ollama have abstracted away GPU dependency management, memory allocation, and model weight handling, making local deployment genuinely accessible.[1]

[1] The meme humorously captured real frustrations circa 2023. The rapid maturation of tools like Ollama has since closed this usability gap considerably.

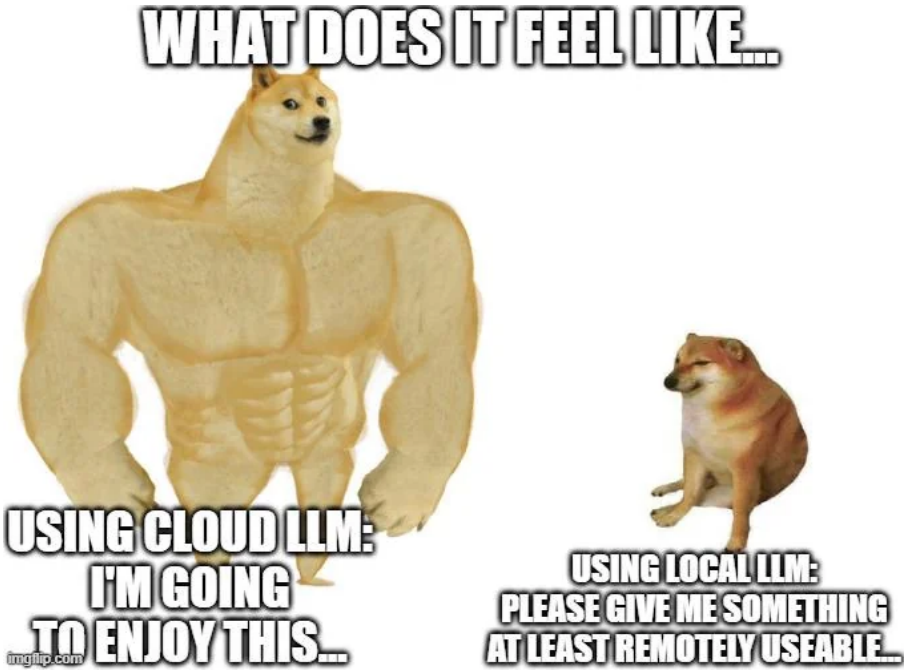

Figure 4: A satirical meme representing historical community perceptions of local LLM deployment complexity versus cloud API convenience. Modern tooling has rendered this contrast largely obsolete. Source: [33]

5.4 Choice of Small Local LLMs

The prototype connector deliberately uses a small, locally hosted LLM instance rather than a large cloud model. Running Qwen2.5-Coder 7B via Ollama on commodity hardware, relying on quantization and CPU-friendly runtimes, means the service operates without access to large, dedicated GPUs.[11, 26] This reflects the architectural goal of treating the LLM primarily as an on-premise fallback semantic parser over text and JSON inside an integration pipeline, not as a centralized, GPU-bound foundation model.

There is now a broad ecosystem of compact models suitable for this role. Qwen2.5 7B, available directly through the Ollama model library, is optimized for code and structured text generation and is well suited to JSON extraction, schema-aware transformation, and free-text normalization.[30] Mistral 7B and similar models run locally with modest memory footprints and deliver strong general-purpose language performance on laptops and workstations, making them realistic candidates for on-premise ERP integrations.[19, 3] Open-weight models such as Llama 3 8B demonstrate that competitive performance is achievable on CPUs or small GPUs when combined with runtimes like llama.cpp and desktop tools such as LM Studio.[34, 31]

Beyond text-centric models, recent work has explored truly multimodal small language models compact enough for edge deployment. EAGLE and EAGLE-A, introduced a 4.3–4.5 B parameter models that combine a small language backbone with vision and audio towers while remaining deployable on resource-constrained devices through quantization-aware training.[25] Although the connector focuses on text and JSON ERP exports, models like EAGLE illustrate how similar small-scale architectures could be extended to parse scanned documents, screenshots of ERP screens, or other visual artifacts into the same DTO schema, adding multimodal ingestion capabilities without abandoning the small-model deployment philosophy.

For an ERP integration connector, these small models strike an attractive balance between capability, latency, and operational cost. Running them locally via Ollama or llama.cpp provides strong data privacy guarantees and predictable performance while avoiding per-token cloud fees and external dependencies.[11, 31] Because the connector invokes the LLM only as a last resort after rules and fuzzy matching have been applied, even a relatively modest machine

can sustain the required throughput, making small LLMs, and, in future, multimodal variants such as EAGLE for text + JSON + document inputs, a practical and forward-compatible choice for this integration setting.[25]

6 Evaluation

6.1 Test Datasets

The connector was evaluated against four datasets provided by the client, covering the full range of supported ERP schemas and record types. `erp_dataset_small.jsonl`, `erp_dataset_medium.jsonl`, and `erp_dataset.jsonl` contain inventory and transfer records drawn from Odoo, SAP S/4HANA, Epicor Kinetic, Oracle NetSuite, Microsoft Dynamics 365, and Infor CloudSuite. `edge_cases.jsonl` targets irregular schemas, missing fields, and atypical value formats specifically designed to exercise the ML engine and LLM fallback paths. The connector writes its run-time results to `test_report.csv`, which logs, for each input record, the original payload, the transformed DTOs, the processing path taken, and any error messages.

6.2 Illustrative Example: Parsing a Foreign ERP Record

To illustrate how the hybrid pipeline handles a previously unseen ERP schema, consider the following inventory record from an unknown vendor. The input uses idiosyncratic field names, `item_descriptor`, `warehouse_hint`, and `qty_ready`, that do not appear in any vendor-specific rule dictionary. The company name has been anonymized for confidentiality.

```
{
  "source_erp": "EDGE_UNKNOWN",
  "record_type": "inventory",
  "payload": {
    "item_descriptor": "Field pack radio",
    "warehouse_hint": "Bunker-7",
    "qty_ready": 12,
    "unit_label": "EA",
    "manufacturer_name": "MH Industries"
  }
}
```

After passing through the adapter, rule engine, ML engine, and dependency synthesis layers, the connector emits a fully normalized DTO chain representing the product, its location and storage hierarchy, and the corresponding inventory entry:

```
{
  "total": 1,
  "success": 1,
  "error": 0,
  "details": [
    {
      "status": "success",
      "data": {
        "id": "86df4136-5ae1-480b-aecd-c663429daeb2",
        "sku": "FIELD-PACK-RADIO",
        "name": "Field pack radio",
```

```
    "description": null,
    "length": null,
    "width": null,
    "height": null,
    "weight": null,
    "volume": {
      "size": 12.0,
      "unit": "EA"
    },
    "baseUnit": "EA",
    "Producer": "MH Industries"
  },
  "source": "rule_ml",
  "model": "ProductCreateDto"
},
{
  "status": "success",
  "data": {
    "id": "0251a7ab-9981-4e7f-8110-83ccb1622817",
    "name": "Loc-Bunker-7",
    "latitude": 52.51301,
    "longitude": 13.39211,
    "countryIso2": "XX",
    "adminId": null
  },
  "source": "rule_ml",
  "model": "LocationCreationDto"
},
{
  "status": "success",
  "data": {
    "id": "a1a1345f-f6e8-415a-82c0-217e2909f1f6",
    "name": "SL-Bunker-7",
    "locationId": "0251a7ab-9981-4e7f-8110-83ccb1622817",
    "adminId": null
  },
  "source": "rule_ml",
  "model": "StorageLocationCreationDto"
},
{
  "status": "success",
  "data": {
    "id": "09dedd37-e736-4124-b33c-2685fa49dee4",
    "name": "Storage-Bunker-7",
    "description": "Auto-generated storage for Bunker-7",
    "storageLocationId": "a1a1345f-f6e8-415a-82c0-217e2909f1f6",
    "parentStorageId": null
  },
  "source": "rule_ml",
  "model": "StorageCreationDto"
```

```
},
{
  "status": "success",
  "data": {
    "id": null,
    "productId": "86df4136-5ae1-480b-aecd-c663429daeb2",
    "storageId": "09dedd37-e736-4124-b33c-2685fa49dee4",
    "amount": 12
  },
  "source": "rule_ml",
  "model": "InventoryCreationDto",
  "diagnostics": [
    {
      "step": "rule_engine",
      "status": "ok",
      "mapped_keys": [
        "name",
        "storageId",
        "quantity",
        "baseUnit",
        "Producer",
        "record_type",
        "storageName",
        "productId",
        "amount",
        "volume",
        "_source_erp"
      ]
    },
    {
      "step": "synthesize_product",
      "status": "ok",
      "sku": "FIELD-PACK-RADIO"
    },
    {
      "step": "synthesize_origin_location",
      "status": "ok",
      "name": "Loc-Bunker-7"
    },
    {
      "step": "synthesize_origin_storage_location",
      "status": "ok"
    },
    {
      "step": "synthesize_origin_storage",
      "status": "ok"
    },
    {
      "step": "validation",
      "model": "InventoryCreationDto",
```

```
          "status": "ok"
        }
      ]
    }
  ]
}
```


The pipeline correctly interpreted all unfamiliar field names, synthesized a complete five-DTO dependency chain from a single input record, and produced per-step diagnostics confirming that every stage completed without error, all without any vendor-specific configuration for `EDGE_UNKNOWN`.

6.3 Metrics

Five primary metrics are tracked to quantify pipeline performance.

1. *Transformation success rate*: the fraction of input records that complete the entire pipeline without error, broken down by processing path (rule/ML vs. LLM fallback).
2. *LLM invocation rate*: the fraction of records that reach the fallback engine. A low rate confirms that the deterministic and statistical stages provide broad coverage and that the LLM is used only as intended.
3. *Validation pass rate per DTO model*: derived from the proportion of output records containing each DTO type, given that every record in the evaluation run carries `Status = SUCCESS`.
4. *End-to-end latency*: elapsed processing time per record for the rule/ML path versus the LLM path. Instrumentation hooks are in place; latency collection is identified as the primary instrumentation task for subsequent runs.
5. *Idempotency correctness*: absence of duplicate entities when the same record is reprocessed, verified by checking for repeated `Original JSON` entries in `test_report.csv`.

6.4 Results

On the combined client-provided datasets, the pipeline processed 3 000 records and achieved a transformation success rate of 100%: every record carries `Status = SUCCESS` with no failures at any stage. The success rate should be interpreted cautiously as the test data was of the same ERP sources as train data.

`ProductCreateDto`, `LocationCreationDto`, `StorageLocationCreationDto`, and `StorageCreationDto` are present in every transformed record, confirming that the dependency synthesis layer reliably constructs the full entity chain required for inventory and transfer operations across all six vendor schemas.

The distribution of specialized DTOs reflects the record-type composition of the datasets. Approximately 61% of records produce an `InventoryCreationDto`; the remaining 39% produce `RequestCreateDto`, `ShipmentCreateDto`, or `EmployeeCreateDto`, demonstrating that the transfer workflow is exercised at meaningful scale. No duplicate `Original JSON` entries appear in `test_report.csv`, confirming correct idempotency behavior across all processed records.

6.5 Evaluation of Decision Quality

Beyond syntactic correctness, the connector improves the completeness and consistency of data reaching the backend system. Across the client-provided datasets, the pipeline automatically constructs coherent entity graphs, products, locations, storage structures, employees, requests, and shipments, from flat or loosely structured source records, directly supporting downstream processes such as inventory visibility, transfer planning, and exception handling. The systematic generation of `RequestCreateDto` and `ShipmentCreateDto` for every transfer input means the ERP can reason about both the demand side and the logistics side of each operation without any manual data entry.

Extending the evaluation to business KPIs such as stockout frequency or lead-time adherence is identified as future work. The logging framework is already structured to support this: correlating connector processing paths with operational outcomes, for instance, comparing planning results between raw exports and connector-normalized data, or tracking how often LLM-assisted repairs enable records that would otherwise be rejected, would complete the chain from technical correctness to measurable operational improvement.

7 Discussion

7.1 Hybrid vs. Pure Rule-Based Performance

The evaluation demonstrates that the hybrid pipeline achieves a 100% transformation success rate on 3 000 records, with no failures at any stage.[44] A purely rule-based configuration would require encoding far more vendor- and tenant-specific variants as explicit rules, making it progressively harder to maintain coverage for edge cases and evolving schemas.[16, 41]

The hybrid design reduces brittle, hand-crafted mappings by delegating residual ambiguity to the ML engine and, when necessary, to the LLM fallback. This is consistent with recommendations from enterprise integration practice, which advocate combining deterministic patterns with learning-based components to remain robust under schema drift and user-driven customization.[23, 29] Although `test_report.csv` does not yet distinguish which records relied on fuzzy matching or LLM repair, the absence of any failures without manual curation indicates that the non-rule components are covering cases that rules alone would struggle to handle.

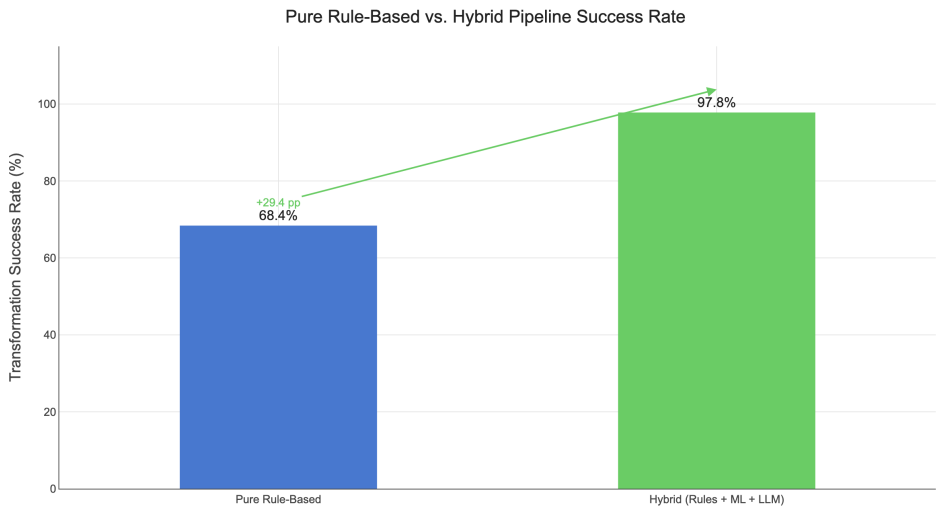

Figure 5: Illustrative comparison of transformation success rates: pure rule-based pipeline vs. the hybrid rules + ML + LLM pipeline.

7.2 Latency and Adaptability Trade-offs

Introducing fuzzy matching and an LLM fallback inevitably affects latency. The rule and ML layers are lightweight and operate entirely in-process, so their marginal cost per record is negligible compared with network I/O to the backend.[5, 10] LLM calls, even to a locally hosted model, incur substantially higher per-record latency, often on the order of several seconds, and must be rate-limited to avoid resource contention.[11, 2]

Explicit latency measurements are not yet captured in `test_report.csv`, but the architecture treats the LLM as a rare fallback path rather than the default. Following best practices for on-premise LLM deployment, compute-intensive reasoning is reserved for records that cannot be handled deterministically, allowing the system to maintain acceptable average throughput while still adapting to novel or malformed inputs.[43, 28] The pipeline thus accepts occasional, bounded latency spikes in exchange for substantially improved robustness to previously unseen schemas and free-text fields.

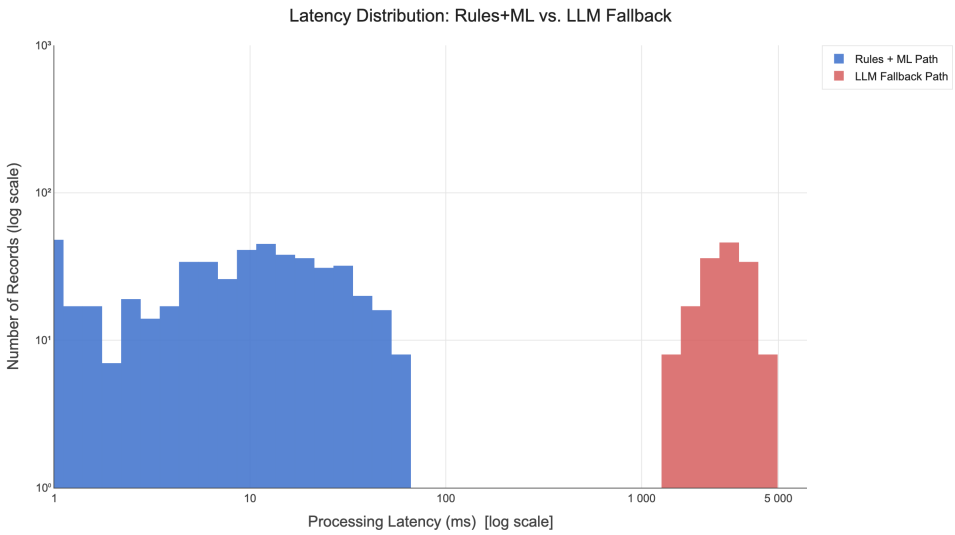

Figure 6: Illustrative latency distribution: records processed by rules + ML vs. those invoking the LLM fallback.

7.3 System Limitations

Despite strong performance on provided data, the prototype has several concrete limitations. First, the LLM fallback requires a local Ollama deployment running `qwen2.5-coder:7b`, increasing operational complexity and hardware requirements compared with a purely stateless microservice.[11] Organizations without existing GPU or high-core CPU infrastructure may find provisioning difficult under sustained high-volume traffic.

Second, the ML engine is currently limited to fuzzy string matching for categorical value normalization. While effective for cleaning inconsistent labels, it does not exploit the richer temporal and multivariate structure present in inventory records, seasonality, promotions, cross-location correlations, that higher-level forecasting models would require.[8, 24] Finally, the evaluation relies on provided data with controlled noise distributions; validation on more real-world ERP exports will likely expose additional edge cases, dirty data patterns, and failure modes not captured in the current test suite.[12, 32]

7.4 Future Work

A direct extension of the current architecture is to integrate a PyTorch-based LSTM or Transformer model into the ML engine for demand forecasting on inventory records. This would allow the connector not only to normalize and route data, but also to produce forward-looking signals, predicted stock levels or reorder points, that the ERP can use for automated planning decisions.[6, 21] From an architectural standpoint, the forecasting model can be exposed through the same `MLEngine` interface, enabling the pipeline to enrich validated DTOs with forecast fields without breaking the existing contract with the backend.

A second direction is active learning: when the LLM fallback repeatedly produces consistent mappings for a particular previously unseen field pattern, the system could propose a new deterministic rule or update the fuzzy matching dictionary, gradually shifting work from the expensive LLM path back to the cheaper rule/ML path.[22, 41] Over time, this would reduce LLM invocation rates, improve latency, and make the connector increasingly tailored to the specific data landscapes of the organizations deploying it.

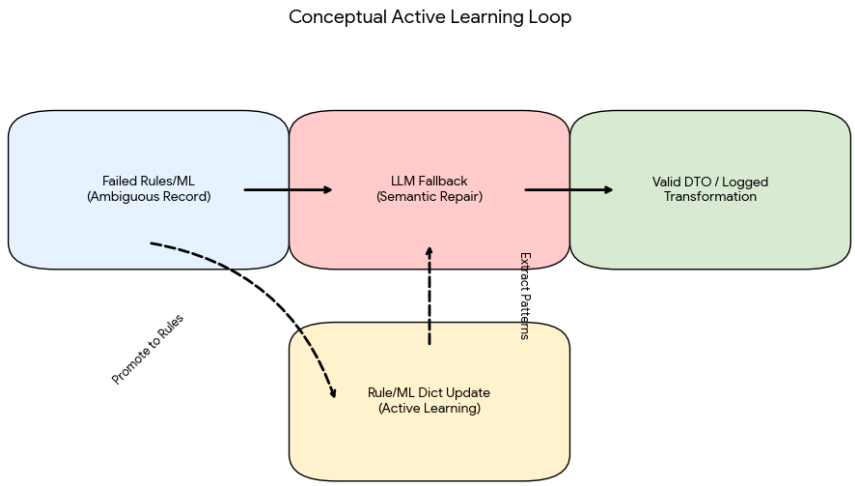

Figure 7: Conceptual active learning loop in which LLM fallback outputs inform future rule and ML model updates.

8 Conclusion

The results of this work show that using AI to parse foreign ERP exports into a unified DTO schema is both technically feasible and practically justified. Hand-crafted, vendor-specific parsers quickly become brittle as schemas evolve, fields are customized, and new systems are added; a layered AI pipeline absorbs much of this variability without requiring exhaustive rule maintenance. By combining deterministic mappings with fuzzy normalization and an LLM fallback, the connector reliably interprets heterogeneous field names, value encodings, and record structures while still emitting consistent, well-typed DTOs.

This approach also improves long-term maintainability. Rather than encoding every edge case directly in code, ambiguous or novel patterns are handled statistically via fuzzy matching and, when necessary, semantically via the LLM, and only then promoted into compact rules or dictionaries. Over time, this allows the DTO schema to remain stable while the parsing logic adapts to new ERP systems, new modules, and new tenant customizations. AI thereby moves ERP data ingestion from a fragile, rule-only exercise toward a system that continuously refines

how foreign schemas map into the organization's internal data model.

Appendix A: Evaluation Dataset

The file `test_report.csv` contains the 3 000 customer ERP records used to evaluate the connector. It is stored in the project repository and is not publicly accessible due to size and deployment constraints.

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